

Customer Segmentation and Purchase Prediction Report

Introduction

This report focuses on customer segmentation and purchase prediction using retail data.

Objectives

Segment customers, identify high-value users, and predict behavior using machine learning.

Dataset Overview

Includes CustomerID, InvoiceNo, Quantity, UnitPrice, and InvoiceDate. TotalPrice was derived.

Data Preprocessing

Removed missing values, filtered invalid data, and aggregated customer-level metrics.

Customer Segmentation

Customers classified into high and low value based on median spending.

EDA Insights

Most customers are low spenders; sales increase over time; strong link between transactions and spending.

Dashboard Insights

4338 customers, 8.89M revenue, strong UK sales dominance, rising monthly trend.

Model Used

Random Forest Classifier using Transactions and Quantity.

Model Performance

Accuracy: 91.82%, Precision: 93.39%, Recall: 90.20%, F1 Score: 91.77%

Conclusion

Machine learning effectively predicts customer segments and supports business decisions.

Metric	Value
Accuracy	91.82%
Precision	93.39%
Recall	90.20%
F1 Score	91.77%